FUJINET • PALM OS

The FujiNet Palm Guide

HotSync, TNFS and network apps for the Palm in your cradle, on FujiNet Classic and FujiNet NIO



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fujinet.online

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CHAPTER 1

Welcome

A Palm Pilot already has everything it needs to be a network computer except the network. It has a serial port, a cradle, and HotSync, the one-button ritual that every Palm owner knows. **FujiNet** supplies the rest. Put a Palm in its serial cradle, run the cradle's cable to a FujiNet, and the same wire now does two jobs:

- **HotSync.** Press the cradle's button and the Palm syncs with FujiNet as it once did with Palm Desktop. FujiNet installs whatever apps and databases are waiting in its install queue, and backs up the Palm's databases to its own storage.
- **FujiBus.** Apps on the Palm talk to FujiNet directly, at 115200 baud, and through it to the internet: game servers, web pages, TNFS file servers, and friends on other Palms.

FujiNet notices which of the two is happening by itself. While an app is talking, it leaves the line alone; when the app goes quiet, it listens now and then for a Palm's HotSync call.

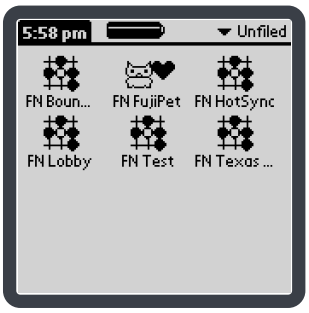
1.1 Two FujiNets, two families of apps

FujiNet comes in two firmware generations, and this guide covers both:

FujiNet Classic is the long-running `fujinet-firmware` that serves Atari, Apple II, CoCo, Adam and many other machines. Its Palm support lives on the `palm-demo-fozz` branch. Apps written for it are named **FN** something, like FN Test. FN

FujiNet NIO is the new firmware, `fujinet-nio`, with devices organised as services (network, files, WiFi, HotSync) and a portable C client library, `fujinet-nio-lib`. Apps written for it are named **FI** something, like FI Test. FI

The two families do the same jobs in the same way on the Palm, and look much alike; the difference is the FujiNet they expect at the other end of the cable. Each FI app has its own creator code, so a Palm can carry both families side by side, and the launcher tells them apart by name.



The FN apps, for FujiNet Classic



The FI apps, for FujiNet NIO

1.2 The apps at a glance

Classic	NIO	Creators	What it does
FN HotSync	FI Config	`ADHS` / `ANHS`	Browse TNFS servers and queue Palm apps for the next HotSync
FN Test	FI Test	`ADTS` / `ANTS`	Check the link to FujiNet, and fetch a web page
FN Bounce World	FI Bounce	`ADBW` / `ANBW`	A shared bouncing-shapes world on a bounce-world server
FN Lobby	FI Lobby	`ADLB` / `ANLB`	Every game server in the FujiNet Game Lobby
FN Texas Hold'em	FI Texas Hold'em	`ADTH` / `ANTH`	Multiplayer poker with Atari, Apple and CoCo players
FN FujiPet	FI FujiPet	`ADFP` / `ANFP`	A virtual pet that meets other Palms' pets online

All twelve run on **Palm OS 3.1 and later**: the Palm III, IIIx, IIIxe, IIIc, V, Vx, VII, VIIx, m100 and their cousins. Chapter 4 has the details.

CHAPTER 2

Classic and NIO

Both firmwares give the Palm the same two services on one cable, HotSync and FujiBus, and both share the line the same way. They differ in how FujiNet is configured, where it keeps the install queue, and what the apps speak on the wire.

2.1 Side by side

	FN Classic	FI NIO
Firmware	`fujinet-firmware`, branch `palm-demo-fozz`; board `fujinet-rs232-s3-palm` (FN-RS232) or FujiNet-PC	`fujinet-nio` with HotSync (pull request #19); the FN-RS232 ESP32-S3 build, or the POSIX build on a computer
Apps	FN *name*, creator `AD..`	FI *name*, creator `AN..`
Talking to FujiNet	FujiBus commands from the app's own small client (`fnlink`); network through the `N1:` device with devicespecs like `N1:TCP://host:port/`	`fujinet-nio-lib`: FujiDevice, NetworkDevice, FileDevice and WiFi services; a lost reply is asked for again and NIO replays it
HotSync settings	`[HotSync]` in `fnconfig.ini`, or the web UI's Palm HotSync panel	`hotsync:` in `fujinet.yaml`, or `hotsync.set` and `hotsync.save` on the console
Install queue	`/palm/install/` on the SD card	`hotsync/install/` on the persistent filesystem (the SD card on an FN-RS232)
Backups	`/palm/backup/`*user*`/`	`hotsync/backup/`*user*`/`
Queueing an app	From the Palm: FN HotSync copies a file from a TNFS host slot. From a computer: WebDAV, `curl -T app.prc http://fujinet/dav/palm/install/`	From the Palm: FI Config copies a file from a TNFS server. From the console: `hotsync.fetch http://host/app.prc`
Network HotSync	TCP 14238, and 6416 for emulators	TCP 14238 (NetSync)
Calendar	Copies a Google or iCalendar calendar into the Date Book at each sync	Not yet
Apps on TNFS	`fujinet.diller.org/PALM/`	`fujinet.diller.org/NIO/PALM/`

2.2 Sharing one cable

A HotSync begins at 9600 baud, while apps talk at 115200. FujiNet cannot hear a 9600-baud call while it is listening at 115200, so both firmwares do the same dance:

1. When no app has spoken for a minute, FujiNet starts *listening windows*: it drops the line to 9600

baud for 1.5 seconds, listens for a Palm's HotSync wake-up, then goes back to 115200.

2. Any app request holds the windows off for another minute, so a running app is never interrupted.
3. Bytes that are not an app request end the wait at once: that is what a pressed HotSync button sounds like at 115200.
4. Once the Palm's wake-up is answered, the sync owns the line until it ends, including the switch to a faster rate.

An app request that lands inside a window gets no answer, so every app resends. On NIO, `fujinet-nio-lib` also wakes the line with a harmless request of its own after an idle spell. You will not notice any of this, except that the first request after a long pause can take a second longer.

> **Tip** A Palm running an app cannot HotSync, and a Palm that is HotSyncing is not running an app. Leave the app (or use its HotSync button) before you press the cradle button.

2.3 Which one do I have?

Run FN Test and FI Test and tap Ping. The one that says FujiNet found! matches your FujiNet. FI Test also reports the board NIO is running on, such as POSIX + FujiBus over RS-232 for a computer running NIO.

CHAPTER 3

Getting set up

3.1 Hardware

- A Palm OS handheld (OS 3.1 or later) and its **serial** HotSync cradle or HotSync cable, the kind with a DB-9 plug. USB cradles do not work.
- A FujiNet with an RS-232 port: the **FN-RS232** board, or a computer running FujiNet-PC or FujiNet NIO with a serial port.
- A **null-modem adapter** between the cradle's DB-9 and the FN-RS232's DB-9. Both ends are wired as a computer would be.

> ⚠ **Watch out** A Palm cradle never raises the RTS/CTS flow-control lines. Build or configure FujiNet without hardware flow control, or it will wait forever for permission to send. The classic `fujinet-rs232-s3-palm` board and NIO's default `flow_control: none` already do.

3.2 FujiNet Classic

Flash the `fujinet-rs232-s3-palm` build of the `palm-demo-fozz` branch. HotSync is on by default; its settings are in `fnconfig.ini` and in the web UI's Palm HotSync panel:

```
[HotSync]
enabled=1
user=FujiNet          ; given to a Palm that has never synced
backup=flagged        ; none | flagged | all
netsync_port=14238    ; 0 turns it off
serial_port=bus       ; the cradle shares the RS-232 bus line
```

An SD card is required: the queue is `/palm/install/`, installed files move to `/palm/installed/`, and backups go to `/palm/backup/`*`user`*`/`. Classic can also fill the Palm's Date Book from a Google or iCalendar calendar at each sync (`calendar=GCAL:///` or `ICAL://host/feed.ics`).

3.3 FujiNet NIO

HotSync is off by default. Turn it on in `fujinet.yaml`, or from the diagnostics console, and restart:

```
hotsync:
  enabled: true
  cradle: true         # a Palm cradle shares the FujiBus serial line
  port: 14238          # NetSync TCP port; 0 turns it off
  root: /hotsync       # folder on the persistent filesystem
  user_name: FujiNet   # given to a Palm that has never synced
  backup: flagged      # none | flagged | all
```

```
hotsync.set enabled 1
hotsync.set cradle 1
hotsync.save
hotsync.status        # is it listening? what did the last sync do?
```

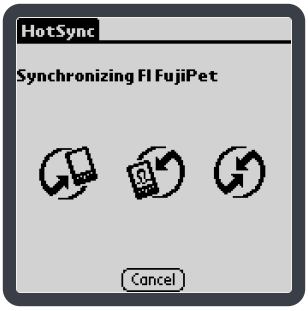
3.4 Installing the apps

The first app has to arrive by HotSync, so put it in the install queue and press the cradle button. After that, FN HotSync or FI Config can queue the rest from the Palm itself. The builds are on the TNFS server `fujinet.diller.org`: the FN apps in `/PALM/`, the FI apps in `/NIO/PALM/`.

Classic Copy the `.prc` to `/palm/install/` on the SD card, or send it over WebDAV:
`curl -T FN-HotSync.prc http://fujinet.local/dav/palm/install/`

NIO Serve the `.prc` from any computer and fetch it from the console:
`python3 -m http.server 8000` (on the computer)
`hotsync.fetch http://192.168.1.5:8000/FI-Config.prc` (on NIO)

Then press the cradle's HotSync button. The Palm shows its HotSync screen, FujiNet installs the queued files and moves them to `installed/`, and the new icons appear in the launcher.



A cradle HotSync with NIO installing a queued app

CHAPTER 4

Palm OS versions

Every app runs on Palm OS 3.1 and later. Palm OS 3.3 brought a new Serial Manager; on 3.1 and 3.2 the apps (and `fujinet-nio-lib`) fall back to the older serial library, which does the same job. An app started on Palm OS 3.0 or earlier says This app needs Palm OS 3.1 or later and quits.

FN FujiPet and FI FujiPet draw in 256 colours on a colour Palm running OS 3.5 or later, and in black and white with a grey dither everywhere else. The other apps are black and white on every Palm.

Palm OS	Devices	Serial	Tested
3.0	Palm III	—	the apps refuse to start
3.1	Palm IIIx, Palm V	serial library	emulator: all twelve apps
3.2	Palm VII	serial library	no ROM available
3.3	Palm III, V (upgraded)	Serial Manager	emulator, and a real Palm in a cradle
3.5	IIIxe, IIIc, Vx, VIIx, m100	Serial Manager	emulator (V, VIIx, m100)
4.x	m500, m505, IIIc	Serial Manager	emulator (IIIc)

Newer Palms Palms from the m500 series onward use a different cradle connector, and their serial cradles are rare. Anything with Palm OS 3.1 or later and a serial cradle works.

CHAPTER 5

The apps

Each app has a page of its own. Where the FN and FI versions differ, the page says how; otherwise everything on it holds for both.

FN Classic	FI NIO	For	Page
FN HotSync		queueing apps from TNFS host slots	9
	FI Config	queueing apps from TNFS servers	10
FN Test	FI Test	checking the cradle link	11
FN Bounce World	FI Bounce	a shared bouncing world	12
FN Lobby	FI Lobby	the FujiNet Game Lobby	13
FN Texas Hold'em	FI Texas Hold'em	multiplayer poker	14
FN FujiPet	FI FujiPet	a virtual pet with friends	15

Finding them New apps land in the launcher's Unfiled category. Pick Unfiled or All from the category list at the top right of the launcher to see them.

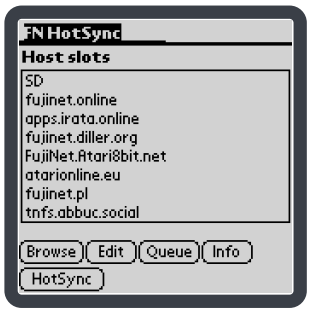
5.1 FN HotSync

FN HotSync

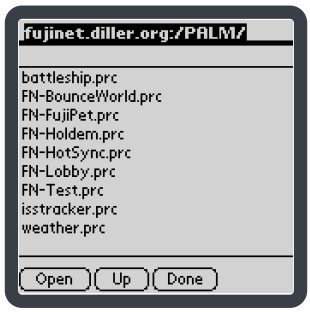
Browse FujiNet's host slots and queue Palm apps for the next HotSync

FN creator ADHS FujiNet Classic

FN HotSync is the Palm's own window on FujiNet Classic's host slots: the same eight slots an Atari or Apple sees in CONFIG, one of them FujiNet's SD card and the rest TNFS servers. Browse a server, pick a Palm app or database, and FN HotSync has FujiNet copy it into `/palm/install/` on the SD card. The next HotSync installs it.



The host slots



Browsing `/PALM/` on a TNFS server



Queue a file for HotSync

Using it

- Pick a host slot and tap Browse. Folders end in `/`: select one and tap Open to go in, Up to come back out.
- Select a `.prc`, `.pdb` or `.pqa` file and tap Open; confirm with Queue. FujiNet copies it to the SD card.
- Queue on the main screen lists what is waiting; HotSync Now there starts the sync.
- Edit names a slot: `SD` for FujiNet's card, or a TNFS server such as `apps.irata.online`. FujiNet saves the slots.
- Info shows FujiNet's WiFi network, IP address and firmware version.
- HotSync closes the cradle port and opens the Palm's HotSync app, ready to sync.

Good to know

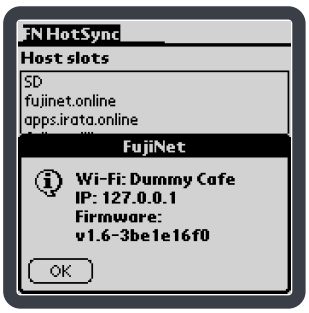
- Queued files go to the slot named `SD`; without one, FN HotSync says so.
- The copy runs on FujiNet, server to SD card, so even a large app takes only a moment.
- A queued file stays until a HotSync installs it; the Palm cannot take it back out of the queue.
- The NIO version of this app is FI Config, on the next page.



Naming a host slot



What is waiting for the next sync



FujiNet's network

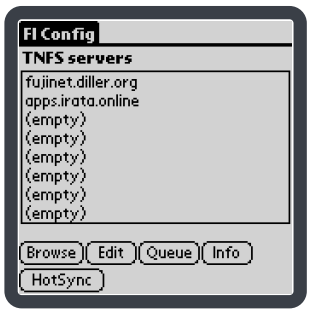
5.2 FI Config

FI Config

Browse TNFS servers and queue Palm apps for HotSync through FujiNet NIO

FI creator ANHS FujiNet NIO

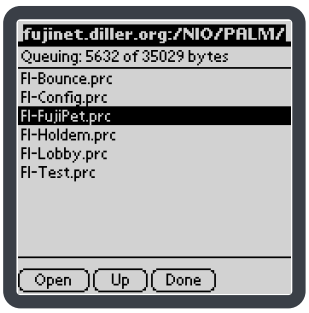
FI Config is FN HotSync rebuilt for FujiNet NIO. NIO has no host slots, so FI Config keeps its own list of eight TNFS servers, starting with `fujinet.diller.org` and `apps.irata.online`. Browse one, pick an app, and FI Config copies it through NIO's file service into `hotsync/install/`, ready for the next HotSync.



The TNFS servers



Browsing a server



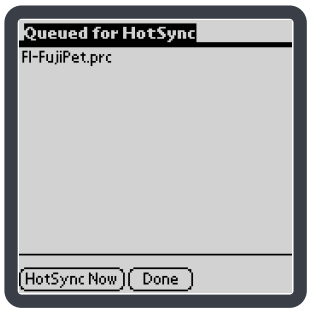
Copying into the queue

Using it

- Pick a server and tap Browse; Open goes into a folder or picks a file, Up goes back.
- Opening a `.prc`, `.pdb` or `.pqa` asks whether to queue it. The copy shows its progress on the line above the list.
- Queue lists the files waiting in NIO's install folder.
- Edit changes a server's name; clear it to empty the slot. The list lives in the app's own preferences.
- Info shows NIO's WiFi network, IP address, firmware and board, and how many replies were asked for again.
- HotSync hands the cradle back and opens the HotSync app.

Good to know

- The Palm does the copying, a few hundred bytes at a time, so a 30 KB app takes about half a minute over TNFS.
- If a copy is cut short, HotSync refuses the half-written file rather than installing a broken app: queue it again to replace it.
- A reply lost on the cable is simply asked for again; Info counts how often that happened.
- On NIO's console, `hotsync.fetch` queues an app over HTTP without the Palm.



The install queue



NIO's firmware and network



The HotSync log: installed

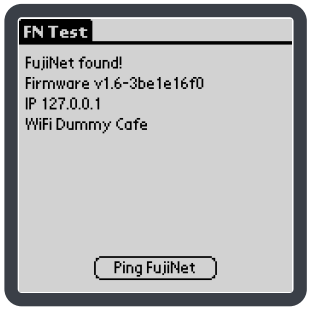
5.3 FN Test and FI Test

FN Test • FI Test

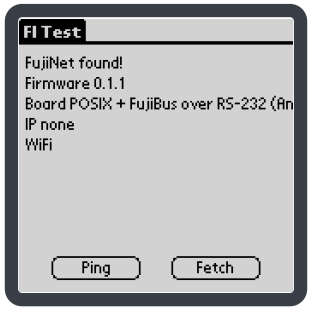
Is FujiNet there? A one-tap check of the cradle link

FN ADTS FI ANTS

The first app to run on a new setup. Ping sends FujiNet one request and shows what comes back. If it says FujiNet found!, the cradle, the cable, the null-modem adapter and the firmware are all doing their job, and every other app will work too.



FN Test finds FujiNet Classic



FI Test finds FujiNet NIO



FI Test fetches a web page

FN Test

Ping FujiNet asks FujiNet Classic for its adapter configuration and shows the firmware version, FujiNet's IP address and the WiFi network it is on.

FI Test

- Ping asks NIO for its version and board (FujiDevice), then for its WiFi status and settings.
- Fetch gets `http://example.com/` through NIO's network service and shows the HTTP status, the size and the page's title: proof that the Palm can reach the internet.

If it fails

No answer from FujiNet means nothing came back after several tries. Check the cable and adapter, and that you are running the right app for your FujiNet: FN Test for Classic, FI Test for NIO. Chapter 2 explains the difference.

> **Tip** Right after a HotSync, or after a long pause, the first ping can take a second longer: FujiNet may be listening for a HotSync at that moment, and the app simply asks again.

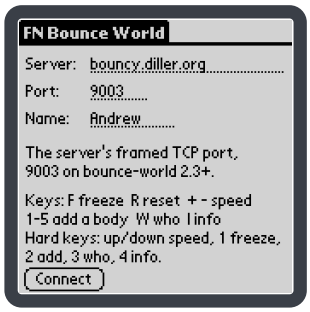
5.4 Bounce World

FN Bounce World • FI Bounce

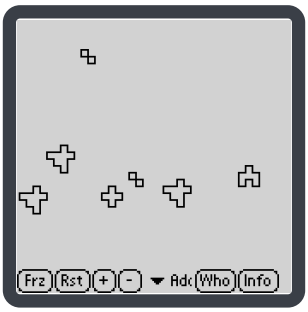
Shapes bouncing across a world shared by every connected machine

FN ADBW FI ANBW

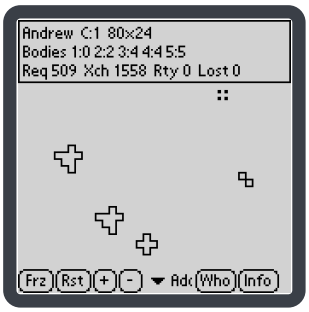
A client for Mark Fisher's bounce-world server. Every machine that connects (an Atari, a C64, a Palm) gets a window onto one shared world, and the shapes in it bounce from screen to screen. The Palm draws its part of the world in block graphics, polling the server several times a second, which also makes it a fine workout for the cradle link.



Connecting to a server



The world in motion



FI Bounce's link counters

Using it

Enter the Server and Port (9003 for the framed protocol of bounce-world 2.3 and later) and a Name of up to 8 letters, then tap Connect. FI Bounce starts with `bouncy.diller.org`; FN Bounce World with `localhost`, for FujiNet-PC.

Frz `F`	freeze or unfreeze the world, for everyone
Rst `R`	reset it
+ -	faster, slower (also the scroll keys)
Add `1`–`5`	add a body of that size
Who `W`	who else is connected
Info `I`	your name, clients, world size and bodies
`Q`	disconnect

While you are in the world the app buttons play too: 1 freezes, 2 adds a body, 3 shows who, 4 shows info.

FI Bounce's counters

FI Bounce's Info adds a line of link statistics: requests, exchanges with NIO, retries and replies lost. A long run with no losses is a healthy cable.

> ⚠ **Known issue** The world keeps drawing while the **World** menu is open, so the menu shows as a blank box. Its items still work, but the buttons and Graffiti keys are easier.

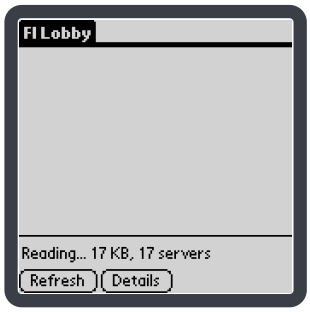
5.5 Lobby

FN Lobby • FI Lobby

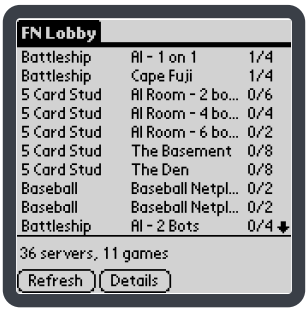
Every game server in the FujiNet Game Lobby, in your pocket

FN ADLB FI ANLB

The FujiNet Game Lobby lists the multiplayer games FujiNet machines can join: Battleship, 5 Card Stud, Texas Hold'em and many more, with who is playing and from which platforms. The Lobby app reads the whole list, `/viewFull` from `lobby.fujinet.online`, and shows it on the Palm.



Reading the lobby



Every server, by game



One server in detail

Using it

- The list loads when the app starts. Each row is a game, a server and its players, such as 0/4.
- Tap a row for the server's status, region, players, app key and URL, and the platforms it serves with the client URL for each. The scroll keys page through it.
- Refresh reads the list again.
- **Lobby > Use QA Lobby** switches to the test lobby, for developers; choose it again to go back.

Good to know

- The reply is about 30 KB of JSON, more than a Palm would like to hold at once. The app reads it as it streams in, keeping only what it shows, up to 48 servers.
- FI Lobby stays usable while the list loads: the menus, Refresh and Details all answer straight away.

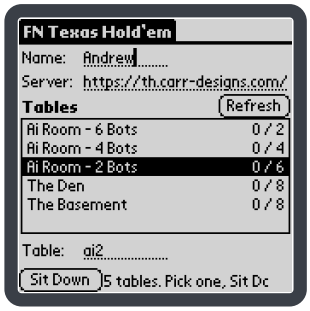
5.6 Texas Hold'em

FN Texas Hold'em • FI Texas Hold'em

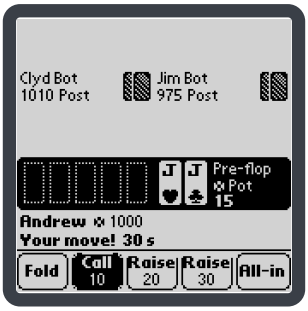
Multiplayer poker, at the same tables as Atari, Apple II and CoCo players

FN ADTH FI ANTH

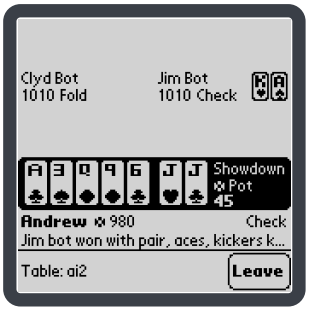
A Palm client for the FujiNet Texas Hold'em server, listed in the Game Lobby. Sit at a table with AI players or with friends on other machines; the server deals and runs every rule, and the Palm shows the felt, the cards and the moves you can make.



Pick a name and a table



Your move



The showdown

Using it

- Enter a Name (up to 8 letters and digits). The tables on the Server are listed with their players; pick one (or type its name under Table) and tap Sit Down.
- Your two cards sit beside the board; the other players are above, face down until the showdown.
- On your turn the moves appear along the bottom with the server's clock. Tap one, or pick one with the scroll keys and press any app button.
- Graffiti works too: `f` fold, `c` check or call, `b` bet, `r` raise, `a` all-in.
- Tap the status line, or **Table > Last Result**, for the last hand. **Table > Leave Table** gives up your seat.

Good to know

- The default server is `th.carr-designs.com`. A lobby address that ends in `?table=den` can be typed as the server, and the table is taken from it.
- The cards are the Apple II client's art, which needs no colour; the small cards are the Atari Lynx client's.
- Leaving the app, or switching to another, leaves the table for you.

5.7 FujiPet

FN FujiPet • FI FujiPet

A virtual pet that grows up like the 1996 original, and makes friends online

FN ADFP FI ANFP

Name your pet and pick a Cat, Dog, Bunny or Dino; the egg hatches five minutes later. It grows from baby to child, teen and adult over a few days, and how well it is looked after decides what it grows into and how long it lives. With Meet friends online on, it keeps a connection to a FujiPet friend server and can chat with pets on other Palms.



A new egg



All grown up, in colour



Friends online

Looking after it

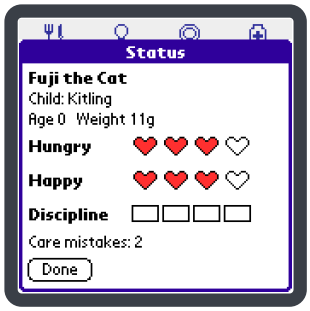
The eight icons are the original's. Along the top: **Feed** (a meal fills a hunger heart, a snack a happy one), **Light**, **Play** (guess which way it will look, three of five to win) and **Medicine**. Along the bottom: **Clean**, **Status**, **Scold** and **Friends**. Tap the pet itself to make it happy. A heart running empty makes it call (the ! in the corner); a call left for 15 minutes is a care mistake. Mess and too many sweets make it sick. The pet lives on while the app is closed, and catches up when you return.

Friends

Register adds your pet to the friend server. The Friends screen lists the pets there, online ones first in bold; pick one, choose one of eight messages under Say and tap Send. It pops up as a bubble on their Palm at once, and a kind one cheers their pet up.

Settings

Options > Settings sets the friend server (host:port), turns going online and sounds on or off, and runs time Fast (a pet minute a second) or Turbo (a pet day a minute) to watch it grow up.



Status: hearts, age and weight



A message from a friend



In greys on a Palm V

CHAPTER 6

When things go wrong

No answer from FujiNet. Is the cradle cabled to it?

The app sent its request several times and heard nothing. Check the null-modem adapter, that FujiNet is running, that it was built without hardware flow control, and that you are running the right family: FN apps with Classic, FI apps with NIO.

This app needs Palm OS 3.1 or later

The Palm runs Palm OS 3.0 or earlier.

The HotSync button does nothing

Quit any FujiNet app first. Check that HotSync is on: on NIO, `hotsync.status` should say `enabled: 1` and `listening: 1`; on Classic, `[HotSync] enabled=1` with `serial_port=bus`. Give it a few seconds: FujiNet listens in windows.

A queued app did not install

Look in the queue (Queue in FN HotSync or FI Config). On NIO, `hotsync.status` reports `last_installed` and `last_install_failures`. A copy that FI Config could not finish is refused by HotSync; queue it again.

FujiPet says No friend server

Check the server's host:port in **Options > Settings**. A computer running NIO on a Mac connects to the IPv6 address for `localhost`, so start the friend server with `--host ""`, or give its IPv4 address.

A FujiNet-PC on a Mac drops requests

FujiNet-PC builds before the `[pc]` serial fix left the terminal's extended input processing on, which swallows bytes `0x16` and `0x0F` in a packet. Use a build with the fix.

APPENDIX A

For developers

A.1 Building

The apps live in `github.com/dillera/fujinet-palm`: the FN apps on the `classic` branch, the FI apps on `nio`. They build with prc-tools-remix and PilRC in a Docker image:

```
make image    # once: the palmdev toolchain image
make          # every app, apps/*/*.prc
make test     # desktop tests of the shared C code
make traps    # Palm OS calls newer than 3.1, per app
```

On `nio`, `make` also builds `fujinet-nio-lib`'s `palm` target from `../fujinet-nio-workspace`. An FI app sets `NIO = 1` in its Makefile; `fn_read` never blocks, so poll it from the event loop, and when a reply is lost ask again at the same offset.

A.2 Testing in an emulator

`assets/` holds Palm OS ROMs and CloudpilotEmu images with Setup done, for Palm OS 3.1, 3.3, 3.5 and 4.0, and the patch that gives CloudpilotEmu a `--serial` option and scripting commands. `tools/rig.sh` joins an emulated Palm's cradle to FujiNet-PC or to FujiNet NIO through a pair of pseudo-terminals:

```
cp assets/images/palm-m100-os3.5.1.img runs/t/palm.img
tools/rig.sh start runs/t runs/t/palm.img nio
tools/rig.sh cmd runs/t install $PWD/apps/FujiNetTest/FujiNetTest.prc
tools/rig.sh cmd runs/t launch 'FI\ Test'
tools/rig.sh cmd runs/t tap 40 147
tools/rig.sh cmd runs/t screenshot ping.png
```

Every screen in this guide was taken that way.